```
clc;
clear;
close all;

% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS

% 1. INPUT PARAMETERS
Fs = 1000;
Fp = 100;
Fst = 150;
Rp = 1;
Rs = 40;

Wp = Fp/(Fs/2);
Wst = Fst/(Fs/2);

% 2. CHEBYSHEV TYPE-I FORMULA
epsilon = sqrt(10^(Rp/10)-1);

N_formula = acosh(sqrt((10^(Rs/10)-1) / ...
    (10^(Rp/10)-1))) / acosh(Fst/Fp);

N_theory = ceil(N_formula);

% 3. BUTTERWORTH FILTER
[Nb,Wb] = buttord(Wp,Wst,Rp,Rs);
[bB,aB] = butter(Nb,Wb,'low');

% 4. CHEBYSHEV TYPE-I FILTER
[Nc,Wc] = cheb1ord(Wp,Wst,Rp,Rs);
[bC,aC] = cheby1(Nc,Rp,Wc,'low');

% 5. ELLIPTIC FILTER
[Ne,We] = ellipord(Wp,Wst,Rp,Rs);
[bE,aE] = ellip(Ne,Rp,Rs,We,'low');

% 6. DISPLAY INPUT PARAMETERS
fprintf('\n==========================================\n');
```

```
==========================================
```

```
fprintf(' IIR FILTER COMPARATIVE ANALYSIS\n');
```

```
 IIR FILTER COMPARATIVE ANALYSIS
```

```
fprintf('==========================================\n');
```

```
==========================================
```

```
fprintf('\nINPUT PARAMETERS\n');
```

```
INPUT PARAMETERS
```

```
fprintf('Sampling Frequency = %.2f Hz\n',Fs);
```

```
Sampling Frequency = 1000.00 Hz
```

```
fprintf('Passband Edge Frequency = %.2f Hz\n',Fp);
```

```
Passband Edge Frequency = 100.00 Hz
```

```
fprintf('Stopband Edge Frequency = %.2f Hz\n',Fst);
```

```
Stopband Edge Frequency = 150.00 Hz
```

```
fprintf('Passband Ripple = %.2f dB\n',Rp);
```

```
Passband Ripple = 1.00 dB
```

```
fprintf('Stopband Attenuation = %.2f dB\n',Rs);
```

```
Stopband Attenuation = 40.00 dB
```

```
% 7. CHEBYSHEV FORMULA RESULTS
fprintf('\nCHEBYSHEV TYPE-I FORMULA\n');
```

```
CHEBYSHEV TYPE-I FORMULA
```

```
fprintf('Ripple factor epsilon = %.4f\n',epsilon);
```

```
Ripple factor epsilon = 0.5088
```

```
fprintf('Calculated order = %.4f\n',N_formula);
```

```
Calculated order = 6.2071
```

```
fprintf('Theoretical minimum N = %d\n',N_theory);
```

```
Theoretical minimum N = 7
```

```
% 8. FILTER ORDER COMPARISON
fprintf('\n==========================================\n');
```

```
==========================================
```

```
fprintf(' FILTER ORDER COMPARISON\n');
```

```
 FILTER ORDER COMPARISON
```

```
fprintf('==========================================\n');
```

```
==========================================
```

```
fprintf('Butterworth Order = %d\n',Nb);
```

```
Butterworth Order = 12
```

```
fprintf('Chebyshev Type-I Order = %d\n',Nc);
```

```
Chebyshev Type-I Order = 6
```

```
fprintf('Elliptic Order = %d\n',Ne);
```

```
Elliptic Order = 4
```

```
% 9. DISPLAY FILTER COEFFICIENTS
fprintf('\n==========================================\n');
```

```
==========================================
```

```
fprintf(' BUTTERWORTH COEFFICIENTS\n');
```

```
 BUTTERWORTH COEFFICIENTS
```

```
fprintf('==========================================\n');
```

```
==========================================
```

```
fprintf('Numerator bB =\n');
```

```
Numerator bB =
```

```
disp(bB);
```

```
   1.0e-03 *

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.
```

```
fprintf('Denominator aB =\n');
```

```
Denominator aB =
```

```
disp(aB);
```

```
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.
```

```
fprintf('\n==========================================\n');
```

```
==========================================
```

```
fprintf(' CHEBYSHEV TYPE-I COEFFICIENTS\n');
```

```
 CHEBYSHEV TYPE-I COEFFICIENTS
```

```
fprintf('==========================================\n');
```

```
==========================================
```

```
fprintf('Numerator bC =\n');
```

```
Numerator bC =
```

```
disp(bC);
```

```
   1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464
```

```
fprintf('Denominator aC =\n');
```

```
Denominator aC =
```

```
disp(aC);
```

```
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586
```

```
fprintf('\n=============================================\n');
```

```

=============================================
```

```
fprintf(' ELLIPTIC COEFFICIENTS\n');
```

```
 ELLIPTIC COEFFICIENTS
```

```
fprintf('=============================================\n');
```

```
=============================================
```

```
fprintf('Numerator bE =\n');
```

```
Numerator bE =
```

```
disp(bE);
```

```
    0.0197   -0.0171    0.0333   -0.0171    0.0197
```

```
fprintf('Denominator aE =\n');
```

```
Denominator aE =
```

```
disp(aE);
```

```
    1.0000   -3.0330    3.8118   -2.2911    0.5554
```

```
% 10. FREQUENCY RESPONSES
N = 4096;

[HB,f] = freqz(bB,aB,N,Fs);
[HC,~] = freqz(bC,aC,N,Fs);
[HE,~] = freqz(bE,aE,N,Fs);

MB = 20*log10(max(abs(HB),1e-12));
MC = 20*log10(max(abs(HC),1e-12));
ME = 20*log10(max(abs(HE),1e-12));

% 11. PASSBAND RIPPLE
P = f <= Fp;
```

```
Ripple_B = max(MB(P))-min(MB(P));
Ripple_C = max(MC(P))-min(MC(P));
Ripple_E = max(ME(P))-min(ME(P));

% 12. STOPBAND ATTENUATION
S = f >= Fst;

Stop_B = -max(MB(S));
Stop_C = -max(MC(S));
Stop_E = -max(ME(S));

% 13. PERCENTAGE DEVIATION
Dev_C_Ripple = abs(Ripple_C-Rp)/Rp*100;
Dev_B_Stop = abs(Stop_B-Rs)/Rs*100;
Dev_C_Stop = abs(Stop_C-Rs)/Rs*100;
Dev_E_Stop = abs(Stop_E-Rs)/Rs*100;

% 14. TRANSITION CHARACTERISTICS
TransitionBW = Fst-Fp;

% 15. DISPLAY PERFORMANCE
fprintf('\n=============================================\n');
```

```
==========================================
```

```
fprintf(' FILTER PERFORMANCE\n');
```

```
 FILTER PERFORMANCE
```

```
fprintf('==========================================\n');
```

```
==========================================
```

```
fprintf('\nPASSBAND RIPPLE\n');
```

```
PASSBAND RIPPLE
```

```
fprintf('Butterworth = %.4f dB\n',Ripple_B);
```

```
Butterworth = 0.8033 dB
```

```
fprintf('Chebyshev Type-I = %.4f dB\n',Ripple_C);
```

```
Chebyshev Type-I = 1.0000 dB
```

```
fprintf('Elliptic = %.4f dB\n',Ripple_E);
```

```
Elliptic = 1.0000 dB
```

```
fprintf('\nSTOPBAND ATTENUATION\n');
```

```
STOPBAND ATTENUATION
```

```
fprintf('Butterworth = %.4f dB\n',Stop_B);
```

```
Butterworth = 40.0198 dB
```

```
fprintf('Chebyshev Type-I = %.4f dB\n',Stop_C);
```

```
Chebyshev Type-I = 41.3365 dB
```

```
fprintf('Elliptic = %.4f dB\n',Stop_E);
```

```
Elliptic = 40.0000 dB
```

```

fprintf('\nCHEBYSHEV RIPPLE DEVIATION\n');
```

```
CHEBYSHEV RIPPLE DEVIATION
```

```
fprintf('Specified Ripple = %.2f dB\n',Rp);
```

```
Specified Ripple = 1.00 dB
```

```
fprintf('Measured Ripple = %.4f dB\n',Ripple_C);
```

```
Measured Ripple = 1.0000 dB
```

```
fprintf('Percentage Deviation = %.4f %%\n',Dev_C_Ripple);
```

```
Percentage Deviation = 0.0000 %
```

```

fprintf('\nSTOPBAND PERCENTAGE DEVIATION\n');
```

```
STOPBAND PERCENTAGE DEVIATION
```

```
fprintf('Butterworth = %.4f %%\n',Dev_B_Stop);
```

```
Butterworth = 0.0494 %
```

```
fprintf('Chebyshev Type-I = %.4f %%\n',Dev_C_Stop);
```

```
Chebyshev Type-I = 3.3413 %
```

```
fprintf('Elliptic = %.4f %%\n',Dev_E_Stop);
```

```
Elliptic = 0.0000 %
```

```

fprintf('\nTRANSITION CHARACTERISTICS\n');
```

```
TRANSITION CHARACTERISTICS
```

```
fprintf('Passband Edge = %.2f Hz\n',Fp);
```

```
Passband Edge = 100.00 Hz
```

```
fprintf('Stopband Edge = %.2f Hz\n',Fst);
```

```
Stopband Edge = 150.00 Hz
```

```
fprintf('Transition Band = %.2f Hz\n',TransitionBW);
```

```
Transition Band = 50.00 Hz
```

```
% 16. MAGNITUDE RESPONSE
figure('Name','IIR Magnitude Response');

plot(f,MB,'LineWidth',1.5);
hold on;
plot(f,MC,'LineWidth',1.5);
plot(f,ME,'LineWidth',1.5);

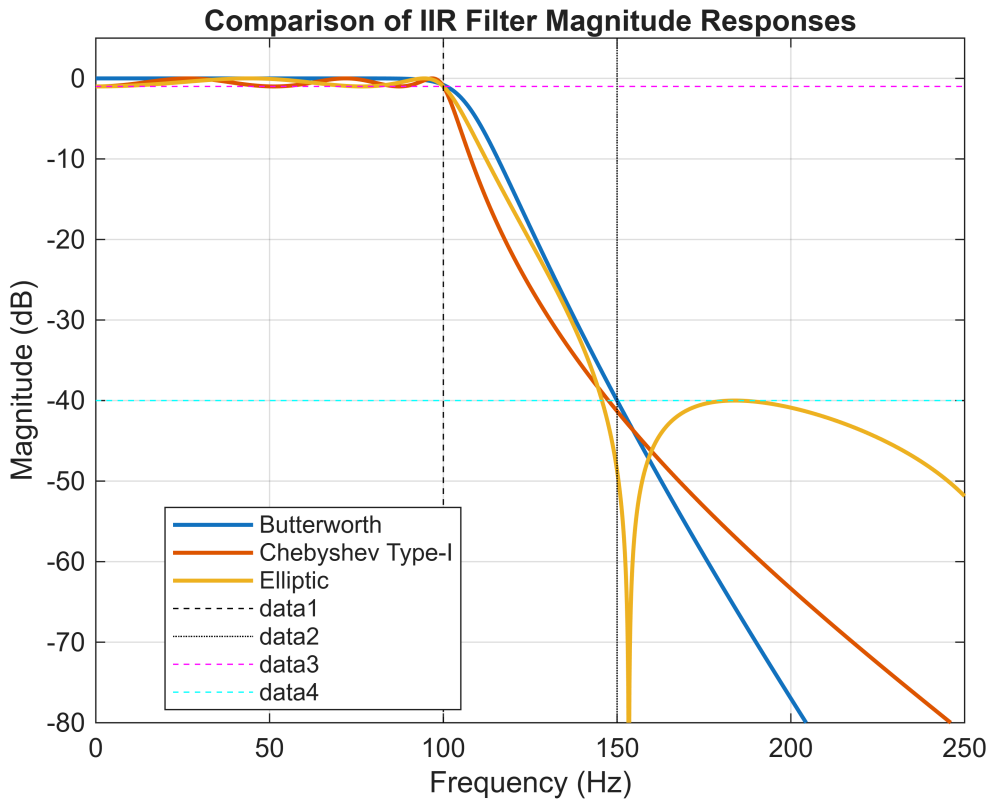
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of IIR Filter Magnitude Responses');

legend('Butterworth','Chebyshev Type-I','Elliptic', ...
    'Location','best');

xlim([0 250]);
ylim([-80 5]);

% Reference lines
plot([Fp Fp],[-80 5],'k--');
plot([Fst Fst],[-80 5],'k:');
plot([0 250],[-Rp -Rp],'m--');
plot([0 250],[-Rs -Rs],'c--');

hold off;
```




```
% 17. PHASE RESPONSE
Phase_B = unwrap(angle(HB))*180/pi;
Phase_C = unwrap(angle(HC))*180/pi;
Phase_E = unwrap(angle(HE))*180/pi;

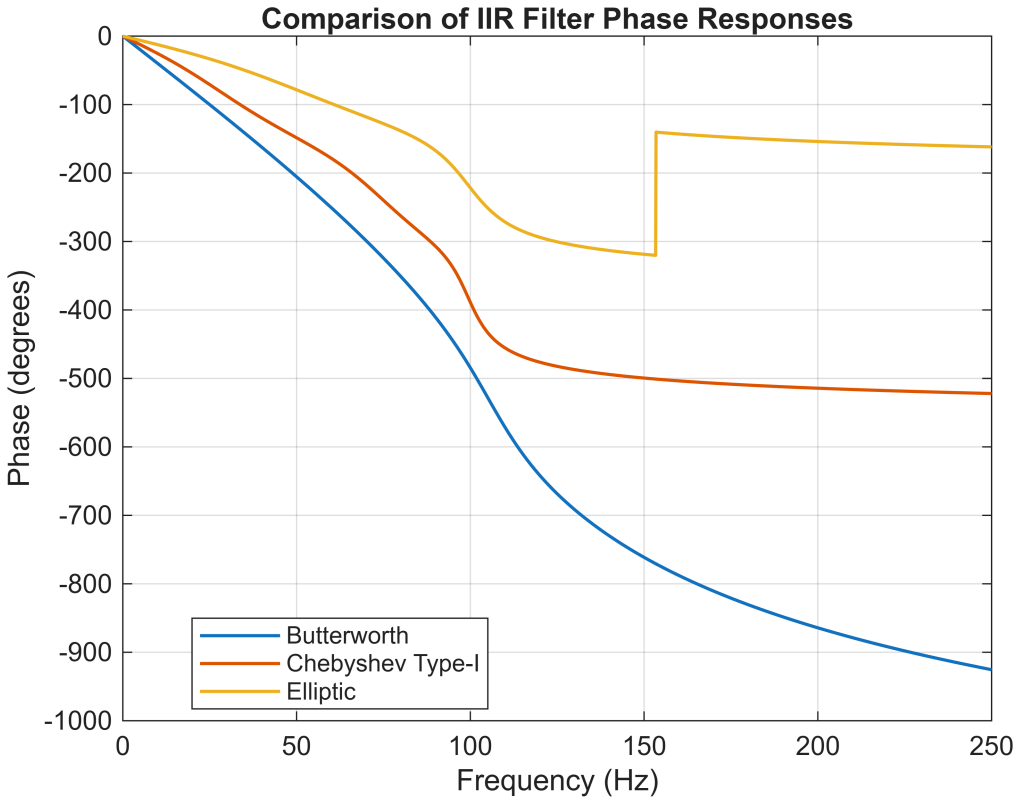
figure('Name','IIR Phase Response');

plot(f,Phase_B,'LineWidth',1.2);
hold on;
plot(f,Phase_C,'LineWidth',1.2);
plot(f,Phase_E,'LineWidth',1.2);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Comparison of IIR Filter Phase Responses');

legend('Butterworth','Chebyshev Type-I','Elliptic', ...
    'Location','best');

xlim([0 250]);
hold off;
```



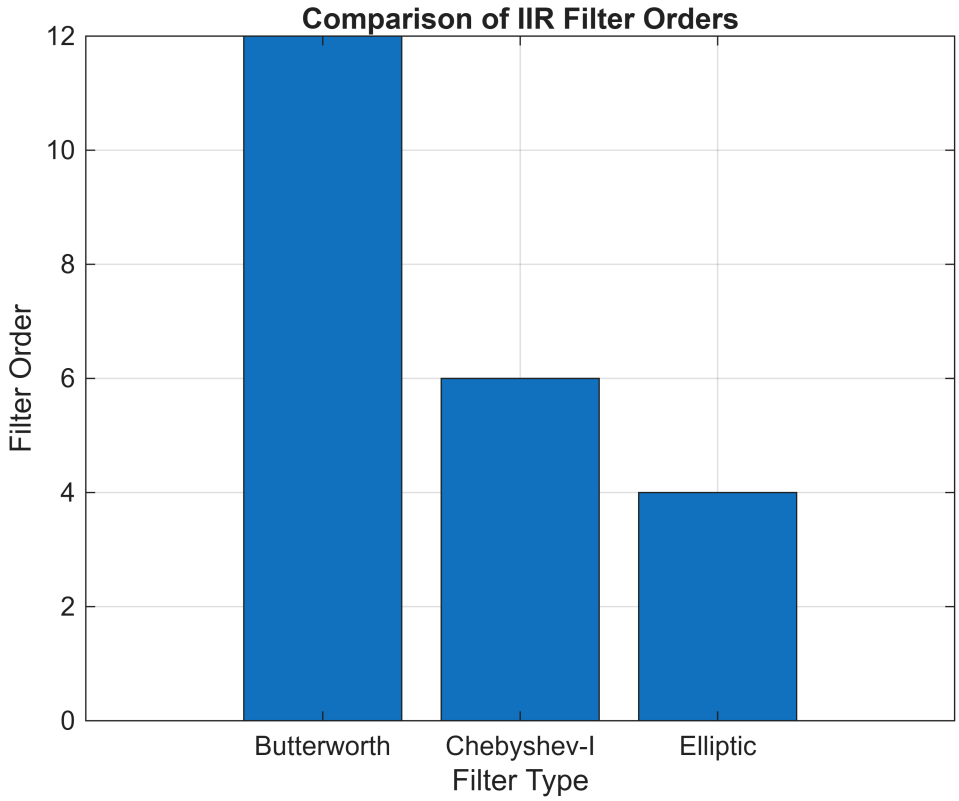

```
% 18. FILTER ORDER GRAPH
figure('Name','IIR Filter Order Comparison');

bar([Nb Nc Ne]);

set(gca,'XTick',1:3);
set(gca,'XTickLabel', ...
    {'Butterworth','Chebyshev-I','Elliptic'});

ylabel('Filter Order');
xlabel('Filter Type');
title('Comparison of IIR Filter Orders');

grid on;
```


Comparison of IIR Filter Orders
Filter Order
0
2
4
6
8
10
12
Butterworth
Chebyshev-I
Elliptic
Filter Type